Machine Learning

Machine learning is a growing technology which enables computers to learn automatically from past data. Machine learning uses various algorithms for **building mathematical models and making predictions using historical data or information**. Currently, it is being used for various tasks such as **image recognition**, **speech recognition**, **email filtering**, **Facebook auto-tagging**, **recommender system**, and many more.

What is Machine Learning

In the real world, we are surrounded by humans who can learn everything from their experiences with their learning capability, and we have computers or machines which work on our instructions. But can a machine also learn from experiences or past data like a human does

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Machine learning enables a machine to automatically learn from data, improve performance from experiences, and predict things without being explicitly programmed.

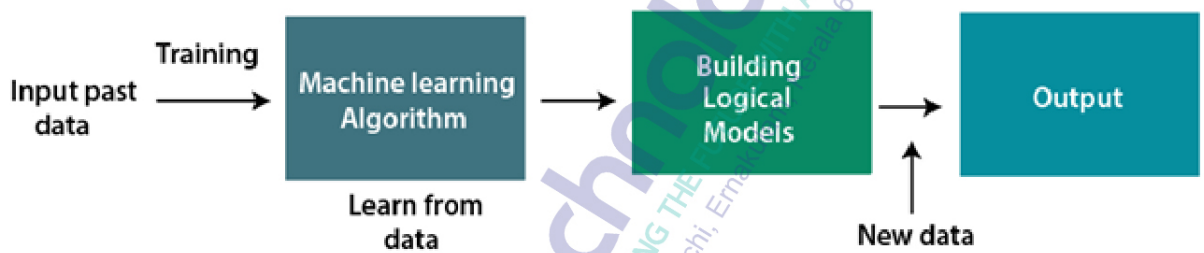
With the help of sample historical data, which is known as **training data**, machine learning algorithms build a **mathematical model** that helps in making predictions or decisions without being explicitly programmed. Machine learning brings computer science and statistics together for creating predictive models. Machine learning constructs or uses the algorithms that learn from historical data. The more we will provide the information, the higher will be the performance.

How does Machine Learning work

A Machine Learning system **learns from historical data, builds the prediction models, and whenever it receives new data, predicts the**

output for it. The accuracy of predicted output depends upon the amount of data, as the huge amount of data helps to build a better model which predicts the output more accurately.

Suppose we have a complex problem, where we need to perform some predictions, so instead of writing a code for it, we just need to feed the data to generic algorithms, and with the help of these algorithms, machine builds the logic as per the data and predict the output. Machine learning has changed our way of thinking about the problem. The below block diagram explains the working of Machine Learning algorithm:



Features of Machine Learning:

- Machine learning uses data to detect various patterns in a given dataset.
- It can learn from past data and improve automatically.
- It is a data-driven technology.
- Machine learning is much similar to data mining as it also deals with the huge amount of the data.

Need for Machine Learning

The need for machine learning is increasing day by day. The reason behind the need for machine learning is that it is capable of doing tasks that are too complex for a person to implement directly. As a human, we have some limitations as we cannot access the huge amount of data manually, so for this, we need some computer systems and here comes the machine learning to make things easy for us.

We can train machine learning algorithms by providing them the huge amount of data and let them explore the data, construct the models, and predict the required output automatically. The performance of the machine learning algorithm depends on the amount of data, and it can be determined by the cost function. With the help of machine learning, we can save both time and money.

The importance of machine learning can be easily understood by its uses cases. Currently, machine learning is used in **self-driving cars**, **cyber fraud detection**, **face recognition**, and **friend suggestion by Facebook**, etc. Various top companies such as Netflix and Amazon have built machine learning models that are using a vast amount of data to analyze the user interest and recommend product accordingly.

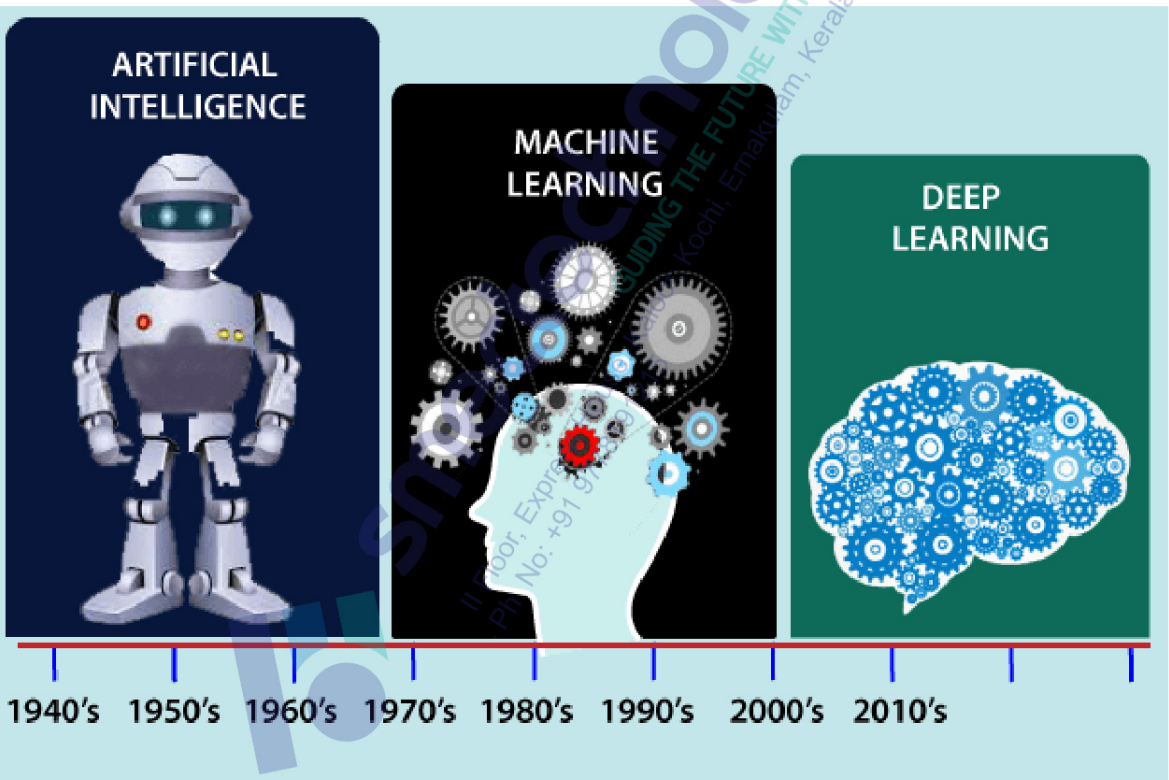
Following are some key points which show the importance of Machine Learning:

- Rapid increment in the production of data
- Solving complex problems, which are difficult for a human
- Decision making in various sector including finance

- Finding hidden patterns and extracting useful information from data.

History of Machine Learning

Before some years (about 40-50 years), machine learning was science fiction, but today it is the part of our daily life. Machine learning is making our day to day life easy from **self-driving cars** to **Amazon virtual assistant "Alexa"**. However, the idea behind machine learning is so old and has a long history. Below some milestones are given which have occurred in the history of machine learning:



The early history of Machine Learning (Pre-1940):

- **1834:** In 1834, Charles Babbage, the father of the computer, conceived a device that could be programmed with punch cards. However, the machine was never built, but all modern computers rely on its logical structure.
- **1936:** In 1936, Alan Turing gave a theory that how a machine can determine and execute a set of instructions.

The era of stored program computers:

- **1940:** In 1940, the first manually operated computer, "ENIAC" was invented, which was the first electronic general-purpose computer. After that stored program computer such as EDSAC in 1949 and EDVAC in 1951 were invented.
- **1943:** In 1943, a human neural network was modeled with an electrical circuit. In 1950, the scientists started applying their idea to work and analyzed how human neurons might work.

Computer machinery and intelligence:

- **1950:** In 1950, Alan Turing published a seminal paper, "**Computer Machinery and Intelligence**," on the topic of artificial intelligence. **In his paper, he asked, "Can machines think?"**

Machine intelligence in Games:

- **1952:** Arthur Samuel, who was the pioneer of machine learning, created a program that helped an IBM computer to play a checkers game. It performed better more it played.

- **1959:** In 1959, the term "Machine Learning" was first coined by **Arthur Samuel**.

The first "AI" winter:

- The duration of 1974 to 1980 was the tough time for AI and ML researchers, and this duration was called as **AI winter**.
- In this duration, failure of machine translation occurred, and people had reduced their interest from AI, which led to reduced funding by the government to the researches.

Machine Learning from theory to reality

- **1959:** In 1959, the first neural network was applied to a real-world problem to remove echoes over phone lines using an adaptive filter.
- **1985:** In 1985, Terry Sejnowski and Charles Rosenberg invented a neural network **NETtalk**, which was able to teach itself how to correctly pronounce 20,000 words in one week.
- **1997:** The IBM's **Deep blue** intelligent computer won the chess game against the chess expert Garry Kasparov, and it became the first computer which had beaten a human chess expert.

Machine Learning at 21st century

- **2006:** In the year 2006, computer scientist Geoffrey Hinton has given a new name to neural net research as "**deep learning**," and nowadays, it has become one of the most trending technologies.

- **2012:** In 2012, Google created a deep neural network which learned to recognize the image of humans and cats in YouTube videos.
- **2014:** In 2014, the Chatbot "**Eugen Goostman**" cleared the Turing Test. It was the first Chabot who convinced the 33% of human judges that it was not a machine.
- **2014: DeepFace** was a deep neural network created by Facebook, and they claimed that it could recognize a person with the same precision as a human can do.
- **2016: AlphaGo** beat the world's number second player **Lee sedol** at **Go game**. In 2017 it beat the number one player of this game **Ke Jie**.
- **2017:** In 2017, the Alphabet's Jigsaw team built an intelligent system that was able to learn the **online trolling**. It used to read millions of comments of different websites to learn to stop online trolling.

Machine Learning at present:

Now machine learning has got a great advancement in its research, and it is present everywhere around us, such as **self-driving cars**, **Amazon Alexa**, **Chatboats**, **recommender system**, and many more. It includes **Supervised**, **unsupervised**, and **reinforcement learning with clustering**, **classification**, **decision tree**, **SVM algorithms**, etc.

Modern machine learning models can be used for making various predictions, including **weather prediction**, **disease prediction**, **stock market analysis**, etc.

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Prerequisites

Before learning machine learning, you must have the basic knowledge of followings so that you can easily understand the concepts of machine learning:

- Fundamental knowledge of probability and linear algebra.
- The ability to code in any computer language, especially in Python language.

Applications of Machine learning

Machine learning is a buzzword for today's technology, and it is growing very rapidly day by day. We are using machine learning in our daily life even without knowing it such as Google Maps, Google assistant, Alexa, etc. Below are some most trending real-world applications of Machine Learning:

1. Image Recognition:

Image recognition is one of the most common applications of machine learning. It is used to identify objects, persons, places, digital images, etc. The popular use case of image recognition and face detection is, **Automatic friend tagging suggestion**:

Facebook provides us a feature of auto friend tagging suggestion. Whenever we upload a photo with our Facebook friends, then we automatically get a tagging suggestion with name, and the technology behind this is machine learning's **face detection** and **recognition algorithm**.

It is based on the Facebook project named "**Deep Face**," which is responsible for face recognition and person identification in the picture.

2. Speech Recognition

While using Google, we get an option of "**Search by voice**," it comes under speech recognition, and it's a popular application of machine learning.

Speech recognition is a process of converting voice instructions into text, and it is also known as "**Speech to text**", or "**Computer speech recognition**." At present, machine learning algorithms are widely used by various applications of speech recognition. **Google assistant**, **Siri**, **Cortana**, and **Alexa** are using speech recognition technology to follow the voice instructions.

3. Traffic prediction:

If we want to visit a new place, we take help of Google Maps, which shows us the correct path with the shortest route and predicts the traffic conditions.

It predicts the traffic conditions such as whether traffic is cleared, slow-moving, or heavily congested with the help of two ways:

- **Real Time location** of the vehicle form Google Map app and sensors
- **Average time has taken** on past days at the same time.

4. Product recommendations:

Machine learning is widely used by various e-commerce and entertainment companies such as **Amazon**, **Netflix**, etc., for product recommendation to

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the user. Whenever we search for some product on Amazon, then we started getting an advertisement for the same product while internet surfing on the same browser and this is because of machine learning.

Google understands the user interest using various machine learning algorithms and suggests the product as per customer interest.

As similar, when we use Netflix, we find some recommendations for entertainment series, movies, etc., and this is also done with the help of machine learning.

5. Self-driving cars:

One of the most exciting applications of machine learning is self-driving cars. Machine learning plays a significant role in self-driving cars. Tesla, the most popular car manufacturing company is working on self-driving car. It is using unsupervised learning method to train the car models to detect people and objects while driving.

6. Email Spam and Malware Filtering:

Whenever we receive a new email, it is filtered automatically as important, normal, and spam. We always receive an important mail in our inbox with the important symbol and spam emails in our spam box, and the technology behind this is Machine learning. Below are some spam filters used by Gmail:

- Content Filter
- Header filter
- General blacklists filter

- Rules-based filters
- Permission filters

7. Virtual Personal Assistant:

We have various virtual personal assistants such as **Google assistant**, **Alexa**, **Cortana**, **Siri**. As the name suggests, they help us in finding the information using our voice instruction. These assistants can help us in various ways just by our voice instructions such as Play music, call someone, Open an email, Scheduling an appointment, etc.

8. Online Fraud Detection:

Machine learning is making our online transaction safe and secure by detecting fraud transaction. Whenever we perform some online transaction, there may be various ways that a fraudulent transaction can take place such as **fake accounts**, **fake ids**, and **steal money** in the middle of a transaction. So to detect this, **Feed Forward Neural network** helps us by checking whether it is a genuine transaction or a fraud transaction.

For each genuine transaction, the output is converted into some hash values, and these values become the input for the next round. For each genuine transaction, there is a specific pattern which gets change for the fraud transaction hence, it detects it and makes our online transactions more secure.

9. Stock Market trading:

Machine learning is widely used in stock market trading. In the stock market, there is always a risk of up and downs in shares, so for this machine learning's **long short term memory neural network** is used for the prediction of stock market trends.

10. Medical Diagnosis:

In medical science, machine learning is used for diseases diagnoses. With this, medical technology is growing very fast and able to build 3D models that can predict the exact position of lesions in the brain.

11. Automatic Language Translation:

Nowadays, if we visit a new place and we are not aware of the language then it is not a problem at all, as for this also machine learning helps us by converting the text into our known languages. Google's GNMT (Google Neural Machine Translation) provide this feature, which is a Neural Machine Learning that translates the text into our familiar language, and it called as automatic translation.

The technology behind the automatic translation is a sequence to sequence learning algorithm, which is used with image recognition and translates the text from one language to another language.

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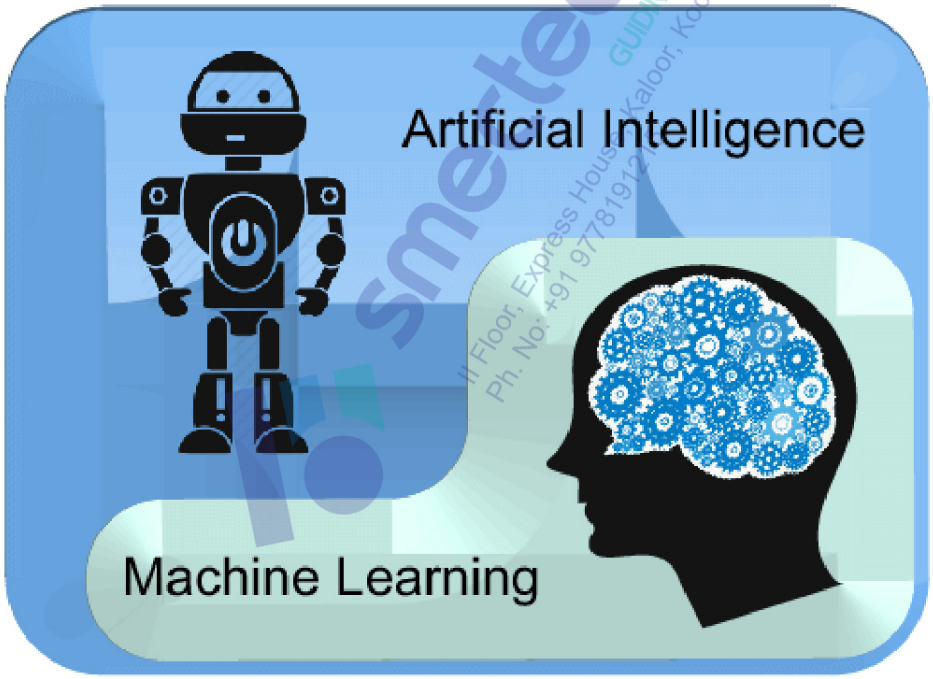
Difference between Artificial intelligence and Machine learning

Artificial intelligence and machine learning are the part of computer science that are correlated with each other. These two technologies are the most trending technologies which are used for creating intelligent systems.

Although these are two related technologies and sometimes people use them as a synonym for each other, but still both are the two different terms in various cases.

On a broad level, we can differentiate both AI and ML as:

AI is a bigger concept to create intelligent machines that can simulate human thinking capability and behavior, whereas, machine learning is an application or subset of AI that allows machines to learn from data without being programmed explicitly.



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Artificial Intelligence

Artificial intelligence is a field of computer science which makes a computer system that can mimic human intelligence. It is comprised of two words "**Artificial**" and "**intelligence**", which means "a human-made thinking power." Hence we can define it as,

Artificial intelligence is a technology using which we can create intelligent systems that can simulate human intelligence.

The Artificial intelligence system does not require to be pre-programmed, instead of that, they use such algorithms which can work with their own intelligence. It involves machine learning algorithms such as Reinforcement learning algorithm and deep learning neural networks. AI is being used in multiple places such as Siri, Google?s AlphaGo, AI in Chess playing, etc.

Key differences between Artificial Intelligence (AI) and Machine learning (ML):

Sl no	Artificial Intelligence	Machine learning
1	Artificial intelligence is a technology which enables a machine to simulate human behavior.	Machine learning is a subset of AI which allows a machine to automatically learn from past data without programming explicitly.
2	The goal of AI is to make a smart computer system like humans to solve complex problems.	The goal of ML is to allow machines to learn from data so that they can give accurate output.
3	In AI, we make intelligent systems to perform any task like a human.	In ML, we teach machines with data to perform a particular task and give an accurate result.

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4	Machine learning and deep learning are the two main subsets of AI.	Deep learning is a main subset of machine learning.
5	AI has a very wide range of scope.	Machine learning has a limited scope.
6	AI system is concerned about maximizing the chances of success.	Machine learning is mainly concerned about accuracy and patterns.
7	The main applications of AI are **Siri, customer support using catboats**, Expert System, Online game playing, intelligent humanoid robot, etc.	The main applications of machine learning are **Online recommender system**, **Google search algorithms**, **Facebook auto friend tagging suggestions**, etc.
8	AI completely deals with Structured, semi-structured, and unstructured data.	Machine learning deals with Structured and semi-structured data.

Machine learning Life cycle

Machine learning has given the computer systems the abilities to automatically learn without being explicitly programmed. But how does a machine learning system work? So, it can be described using the life cycle of machine learning. Machine learning life cycle is a cyclic process to build an efficient machine learning project. The main purpose of the life cycle is to find a solution to the problem or project.

Machine learning life cycle involves seven major steps, which are given below:

1. ***Gathering Data***
2. ***Data preparation***

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3. *Data Wrangling*
4. *Analyse Data*
5. *Train the model*
6. *Test the model*
7. *Deployment*

The most important thing in the complete process is to understand the problem and to know the purpose of the problem. Therefore, before starting the life cycle, we need to understand the problem because the good result depends on the better understanding of the problem.

In the complete life cycle process, to solve a problem, we create a machine learning system called "model", and this model is created by providing "training". But to train a model, we need data, hence, life cycle starts by collecting data.

1. Gathering Data:

Data Gathering is the first step of the machine learning life cycle. The goal of this step is to identify and obtain all data-related problems.

In this step, we need to identify the different data sources, as data can be collected from various sources such as **files**, **database**, **internet**, or **mobile devices**. It is one of the most important steps of the life cycle. The quantity and quality of the collected data will determine the efficiency of the output. The more will be the data, the more accurate will be the prediction.

This step includes the below tasks:

- **Identify various data sources**

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- **Collect data**
- **Integrate the data obtained from different sources**

2. Data preparation

After collecting the data, we need to prepare it for further steps. Data preparation is a step where we put our data into a suitable place and prepare it to use in our machine learning training.

In this step, first, we put all data together, and then randomize the ordering of data.

This step can be further divided into two processes:

- **Data exploration:**
  It is used to understand the nature of data that we have to work with. We need to understand the characteristics, format, and quality of data.A better understanding of data leads to an effective outcome. In this, we find Correlations, general trends, and outliers.

- **Data pre-processing:**
  Now the next step is preprocessing of data for its analysis.

3. Data Wrangling

Data wrangling is the process of cleaning and converting raw data into a useable format. It is the process of cleaning the data, selecting the variable to use, and transforming the data in a proper format to make it more suitable for analysis in the next step. It is one of the most important steps of the complete process. Cleaning of data is required to address the quality issues.

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It is not necessary that data we have collected is always of our use as some of the data may not be useful. In real-world applications, collected data may have various issues, including:

- **Missing Values**
- **Duplicate data**
- **Invalid data**
- **Noise**

So, we use various filtering techniques to clean the data.

4. Data Analysis

Now the cleaned and prepared data is passed on to the analysis step. This step involves:

- **Selection of analytical techniques**
- **Building models**
- **Review the result**

The aim of this step is to build a machine learning model to analyze the data using various analytical techniques and review the outcome. It starts with the determination of the type of the problems, where we select the machine learning techniques such as **Classification**, **Regression**, **Cluster analysis**, **Association**, etc. then build the model using prepared data, and evaluate the model.

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5. Train Model

Now the next step is to train the model, in this step we train our model to improve its performance for better outcome of the problem.

We use datasets to train the model using various machine learning algorithms. Training a model is required so that it can understand the various patterns, rules, and, features.

6. Test Model

Once our machine learning model has been trained on a given dataset, then we test the model. In this step, we check for the accuracy of our model by providing a test dataset to it.

Testing the model determines the percentage accuracy of the model as per the requirement of project or problem.

7. Deployment

The last step of machine learning life cycle is deployment, where we deploy the model in the real-world system.

If the above-prepared model is producing an accurate result as per our requirement with acceptable speed, then we deploy the model in the real system. But before deploying the project, we will check whether it is improving its performance using available data or not. The deployment phase is similar to making the final report for a project.

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How to get datasets for Machine Learning

The key to success in the field of machine learning or to become a great data scientist is to practice with different types of datasets. But discovering a suitable dataset for each kind of machine learning project is a difficult task.

What is a dataset?

A dataset is a collection of data in which data is arranged in some order. A dataset can contain any data from a series of an array to a database table. Below table shows an example of the dataset:

Country	Age	Salary	Purchased
India	38	48000	No
France	43	45000	Yes
Germany	30	54000	No
France	48	65000	No
Germany	40		Yes
India	35	58000	Yes

A tabular dataset can be understood as a database table or matrix, where each column corresponds to a **particular variable,** and each row corresponds to the **fields of the dataset.** The most supported file type for a tabular dataset is **"Comma Separated File,"** or **CSV.** But to store a "tree-like data," we can use the JSON file more efficiently.

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Types of data in datasets

- **Numerical data:**Such as house price, temperature, etc.
- **Categorical data:**Such as Yes/No, True/False, Blue/green, etc.
- **Ordinal data:**These data are similar to categorical data but can be measured on the basis of comparison.

Need of Dataset

To work with machine learning projects, we need a huge amount of data, because, without the data, one cannot train ML/AI models. Collecting and preparing the dataset is one of the most crucial parts while creating an ML/AI project.

The technology applied behind any ML projects cannot work properly if the dataset is not well prepared and pre-processed.

During the development of the ML project, the developers completely rely on the datasets. In building ML applications, datasets are divided into two parts:

- **Training dataset:**
- **Test Dataset**

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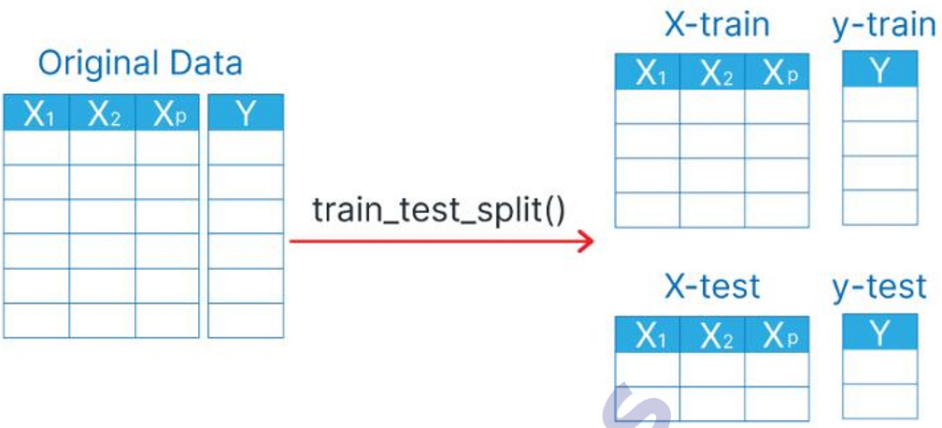

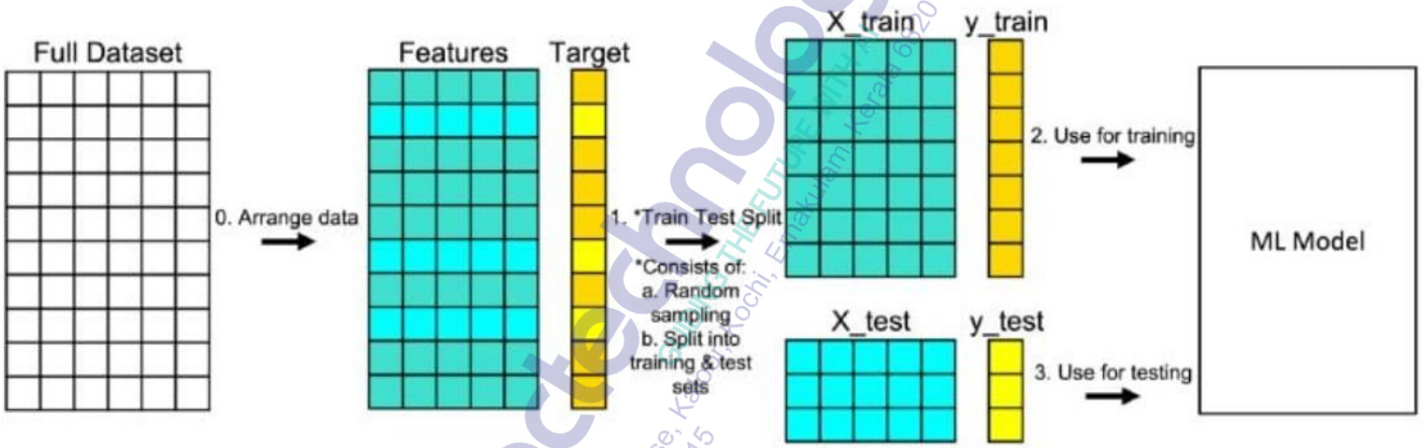

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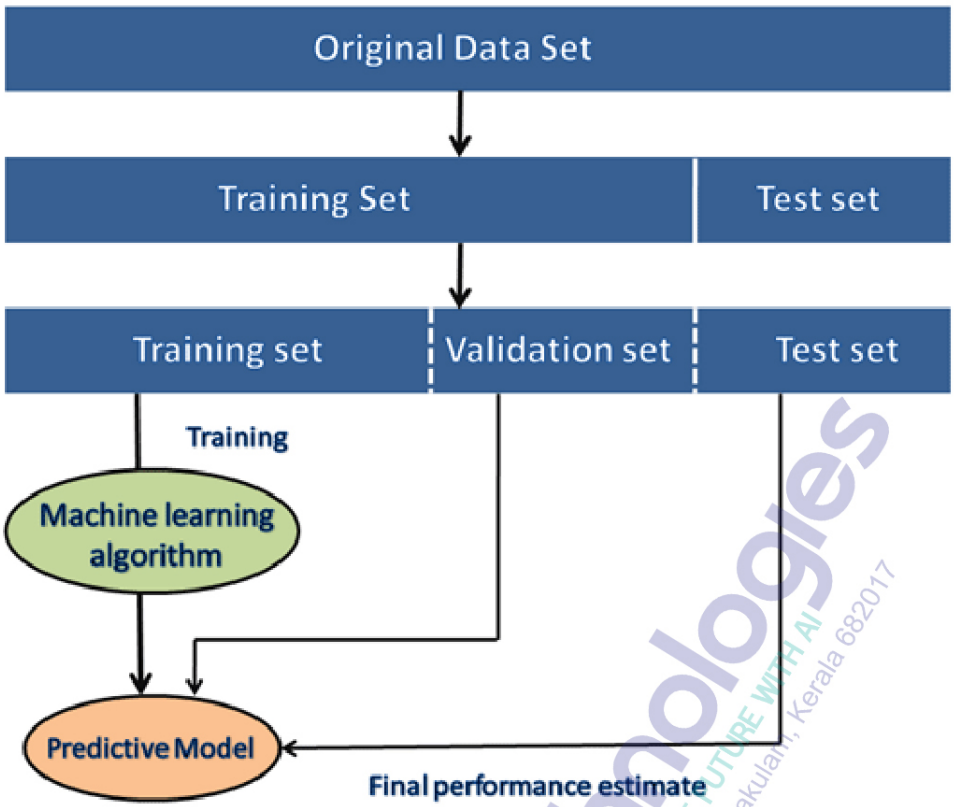

Popular sources for Machine Learning datasets

Below is the list of datasets which are freely available for the public to work on it:

1. Kaggle Datasets

Kaggle is one of the best sources for providing datasets for Data Scientists and Machine Learners. It allows users to find, download, and publish datasets in an easy way. It also provides the opportunity to work with other machine learning engineers and solve difficult Data Science related tasks.

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Kaggle provides a high-quality dataset in different formats that we can easily find and download.

https://www.kaggle.com/datasets

2. UCI Machine Learning Repository

UCI Machine learning repository is one of the great sources of machine learning datasets. This repository contains databases, domain theories, and data generators that are widely used by the machine learning community for the analysis of ML algorithms.

The link for the UCI machine learning repository is https://archive.ics.uci.edu/ml/index.php

3. Datasets via AWS

We can search, download, access, and share the datasets that are publicly available via AWS resources. These datasets can be accessed through AWS resources but provided and maintained by different government organizations, researches, businesses, or individuals.

Anyone can analyze and build various services using shared data via AWS resources. The shared dataset on cloud helps users to spend more time on data analysis rather than on acquisitions of data.

This source provides the various types of datasets with examples and ways to use the dataset. It also provides the search box using which we can search for the required dataset. Anyone can add any dataset or example to the **Registry of Open Data on AWS.**

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The link for the resource is

https://registry.opendata.aws/

4. Google's Dataset Search Engine

Google dataset search engine is a search engine launched by **Google** on **September 5, 2018.** This source helps researchers to get online datasets that are freely available for use.

The link for the Google dataset search engine is

https://datasetsearch.research.google.com/

5. Microsoft Datasets

The Microsoft has launched the **"Microsoft Research Open data"** repository with the collection of free datasets in various areas such as **natural language processing, computer vision, and domain-specific sciences.**

Using this resource, we can download the datasets to use on the current device, or we can also directly use it on the cloud infrastructure.

The link to download or use the dataset from this resource is

https://msropendata.com/

6. Awesome Public Dataset Collection

Awesome public dataset collection provides high-quality datasets that are arranged in a well-organized manner within a list according to topics such as Agriculture, Biology, Climate, Complex networks, etc. Most of the datasets

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are available free, but some may not, so it is better to check the license before downloading the dataset.

The link to download the dataset from Awesome public dataset collection is

https://github.com/awesomedata/awesome-public-datasets

7. Government Datasets

There are different sources to get government-related data. Various countries publish government data for public use collected by them from different departments.

The goal of providing these datasets is to increase transparency of government work among the people and to use the data in an innovative approach. Below are some links of government datasets:

- Indian Government dataset

https://data.gov.in/

- US Government Dataset

https://data.gov/

- Northern Ireland Public Sector Datasets

https://www.opendatani.gov.uk/

- European Union Open Data Portal

https://data.europa.eu/en

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8. Computer Vision Datasets

Visual data provides multiple numbers of the great dataset that are specific to computer visions such as Image Classification, Video classification, Image Segmentation, etc. Therefore, if you want to build a project on deep learning or image processing, then you can refer to this source.

The link for downloading the dataset from this source is https://www.visualdata.io/.

9. Scikit-learn dataset

Scikit-learn is a great source for machine learning enthusiasts. This source provides both toy and real-world datasets. These datasets can be obtained from sklearn.datasets package and using general dataset API.

The toy dataset available on scikit-learn can be loaded using some predefined functions such as, **load_boston([return_X_y]), load_iris([return_X_y]),** etc, rather than importing any file from external sources. But these datasets are not suitable for real-world projects.

The link to download datasets from this source is

https://scikit-learn.org/stable/datasets

Data Preprocessing in Machine learning

Data preprocessing is a process of preparing the raw data and making it suitable for a machine learning model. It is the first and crucial step while creating a machine learning model.

When creating a machine learning project, it is not always a case that we come across the clean and formatted data. And while doing any operation with data, it is mandatory to clean it and put in a formatted way. So for this, we use data preprocessing task.

Why do we need Data Preprocessing?

A real-world data generally contains noises, missing values, and maybe in an unusable format which cannot be directly used for machine learning models. Data preprocessing is required tasks for cleaning the data and making it suitable for a machine learning model which also increases the accuracy and efficiency of a machine learning model.

- *Getting the dataset*
- *Importing libraries*
- *Importing datasets*
- *Finding Missing Data*
- *Encoding Categorical Data*
- *Splitting dataset into training and test set*
- *Feature scaling*

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1) Get the Dataset

To create a machine learning model, the first thing we required is a dataset as a machine learning model completely works on data. The collected data for a particular problem in a proper format is known as the **dataset**.

Dataset may be of different formats for different purposes, such as, if we want to create a machine learning model for business purpose, then dataset

will be different with the dataset required for a liver patient. So each dataset is different from another dataset. To use the dataset in our code, we usually put it into a CSV **file**. However, sometimes, we may also need to use an HTML or xlsx file.

What is a CSV File?

CSV stands for "**Comma-Separated Values**" files; it is a file format which allows us to save the tabular data, such as spreadsheets. It is useful for huge datasets and can use these datasets in programs.

2) Importing Libraries

In order to perform data preprocessing using Python, we need to import some predefined Python libraries. These libraries are used to perform some specific jobs. There are three specific libraries that we will use for data preprocessing, which are:

Numpy: Numpy Python library is used for including any type of mathematical operation in the code. It is the fundamental package for scientific calculation in Python. It also supports to add large, multidimensional arrays and matrices. So, in Python, we can import it as:

```
import numpy as np
```

Matplotlib: The second library is **matplotlib**, which is a Python 2D plotting library, and with this library, we need to import a sub-library **pyplot**. This library is used to plot any type of charts in Python for the code. It will be imported as below:

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```
import matplotlib.pyplot as plt
```

Pandas: The last library is the Pandas library, which is one of the most famous Python libraries and used for importing and managing the datasets. It is an open-source data manipulation and analysis library. It will be imported as below:

```
import pandas as pd
```

3) Importing the Datasets

Now we need to import the datasets which we have collected for our machine learning project. But before importing a dataset, we need to set the current directory as a working directory.

```
import pandas as pd

df=pd.read_csv(r"C:\DataScience\Dataset\ml\Data_2.csv")
# The command print(df.to_string()) in Python is used  to
# print the entire contents of a pandas DataFrame df as a
# formatted string. Unlike the default print(df), which may
# truncate the output for large DataFrames, to_string()
# ensures that all rows and columns are displayed in the
# console output.
print(df.to_string())
```

Output:

```
       Country       Age    Salary           Purchased
0      France        44.0   72000.0              No
```

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1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40.0	NaN	Yes
5	France	35.0	58000.0	Yes
6	Spain	NaN	52000.0	No
7	France	48.0	79000.0	Yes
8	Germany	50.0	83000.0	No
9	France	37.0	67000.0	Yes

Dependent variables are nothing but the variable which holds the phenomena which we are studying. Independent variables are the ones which through we are trying to explain the value or effect of the output variable (dependent variable) by creating a relationship between an independent and dependent variable.

Features in machine learning

In machine learning, **features** (also known as **predictors** or **independent variables**) are individual measurable properties or characteristics of the data used to make predictions or understand patterns. The selection, transformation, and engineering of features play a critical role in the effectiveness of a machine learning model.

1. Numerical Features

- **Description:** These are features that represent quantities and are typically continuous (can take any value within a range) or discrete (take integer values).
- **Examples:**

- **Continuous**: Age, income, temperature, height, weight.
- **Discrete**: Number of children, number of products sold.

- **Use cases:** Regression problems, such as predicting house prices or sales amounts.

2. Categorical Features

- **Description:** These are features that represent categories or labels. The values are typically discrete and finite but are not ordered or measurable.
- **Examples:**
    - Gender (Male, Female)
    - City (New York, London, Tokyo)
    - Product category (Electronics, Clothing, Furniture)
- **Use cases:** Classification problems, such as determining if an email is spam or not.

3. Ordinal Features

- **Description:** These are features that have categories with a meaningful order or ranking, but the differences between the categories are not well-defined.
- **Examples:**
    - Education level (High School, Bachelor's, Master's, PhD)
    - Customer satisfaction (Very Poor, Poor, Neutral, Good, Excellent)

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- T-shirt sizes (Small, Medium, Large, Extra Large)

- **Use cases:** Ranking or classification problems where the order matters but not the exact difference between the categories.

4. Binary Features

- **Description:** These are categorical features with only two possible values, often used to indicate the presence or absence of a certain property.
- **Examples:**
    - Is_Active (Yes/No)
    - Has_Pets (True/False)
    - Is_Employed (1/0)
- **Use cases:** Binary classification problems, such as predicting whether a customer will churn (Yes/No).

5. Date and Time Features

- **Description:** These are features that relate to specific points in time, such as dates and timestamps. Time-related features can be transformed into other useful features like year, month, weekday, or hour.
- **Examples:**
    - Timestamp of a transaction
    - Day of the week, month, quarter, or year
    - Hour of the day, season (summer, fall)

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- **Use cases:** Time series forecasting (e.g., stock price prediction), seasonal effects in sales prediction.

6. Text Features

- **Description:** These are features extracted from unstructured text data. Text data can be converted into numerical features using techniques like **TF-IDF**, **word embeddings**, or **Bag of Words**.
- **Examples:**
  - Customer reviews (sentiment analysis)
  - Emails (spam classification)
  - Tweets (topic modeling)
- **Use cases:** Natural language processing tasks like sentiment analysis, text classification, and machine translation.

7. Image Features

- **Description:** In image data, features can be pixel values, patterns, or objects detected in an image. Deep learning models such as Convolutional Neural Networks (CNNs) are typically used to extract meaningful features from images.
- **Examples:**
  - Pixel intensities in an image
  - Features from an image object detection algorithm (e.g., the presence of a cat, dog, or car)
- **Use cases:** Image classification, object detection, and segmentation tasks.

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8. Derived or Engineered Features

- **Description:** These are features created from existing data through mathematical transformations, domain knowledge, or combinations of multiple features. Feature engineering is a key part of improving machine learning models.
- **Examples:**
  - Combining height and weight to create a **Body Mass Index (BMI)** feature
  - Using the ratio of income to debt to create a **Debt-to-Income ratio**
  - Extracting **mean**, **median**, or **standard deviation** from time series data
- **Use cases:** Feature engineering is done to create more meaningful features that help improve model accuracy.

9. Interaction Features

- **Description:** These are features created by combining two or more existing features to capture interactions between them. Sometimes, the interaction between features may provide more valuable insights than individual features alone.
- **Examples:**
  - Combining **age** and **income** to create a new feature like **age-income group**.
  - Multiplying or adding two numerical features (e.g., price * quantity).

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- **Use cases:** Interaction terms in regression models (e.g., capturing non-linear effects between features).

Feature Selection

Identify the best features or the dominating variable that could be used to predict an outcome

1. Feature selection using correlation

Correlation measures the linear relationship between two variables. You can use the correlation matrix to select features that are highly correlated with the target variable and avoid features that are highly correlated with each other.

- Highly correlated features might lead to multicollinearity, so it's better to drop one of the features.
- Target correlation: Features with high correlation to the target variable are usually good candidates

Multicollinearity means if two column are correlated ie as one variable changes other variable also changes which can be increasing or decreasing. If the variable changes that means they are correlated. Multicollinearity means if this kind of variables exist you have to remove any of the two, otherwise this leads to confusion in the output predictions.

```
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

# Load the Iris dataset
```

```
df = sns.load_dataset('iris')

# Convert the 'species' column to numeric using factorize
df['species'] = pd.factorize(df['species'])[0]

# Compute the correlation matrix
correlation_matrix = df.corr()

# Plot the correlation matrix heatmap
plt.figure(figsize=(8, 6))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm',
fmt='.2f')
plt.title("Correlation Matrix of Iris Dataset")
plt.show()

# Select features that are highly correlated with the target
(species)
species_corr = correlation_matrix['species']
print("Correlation with species:\n", species_corr)
```

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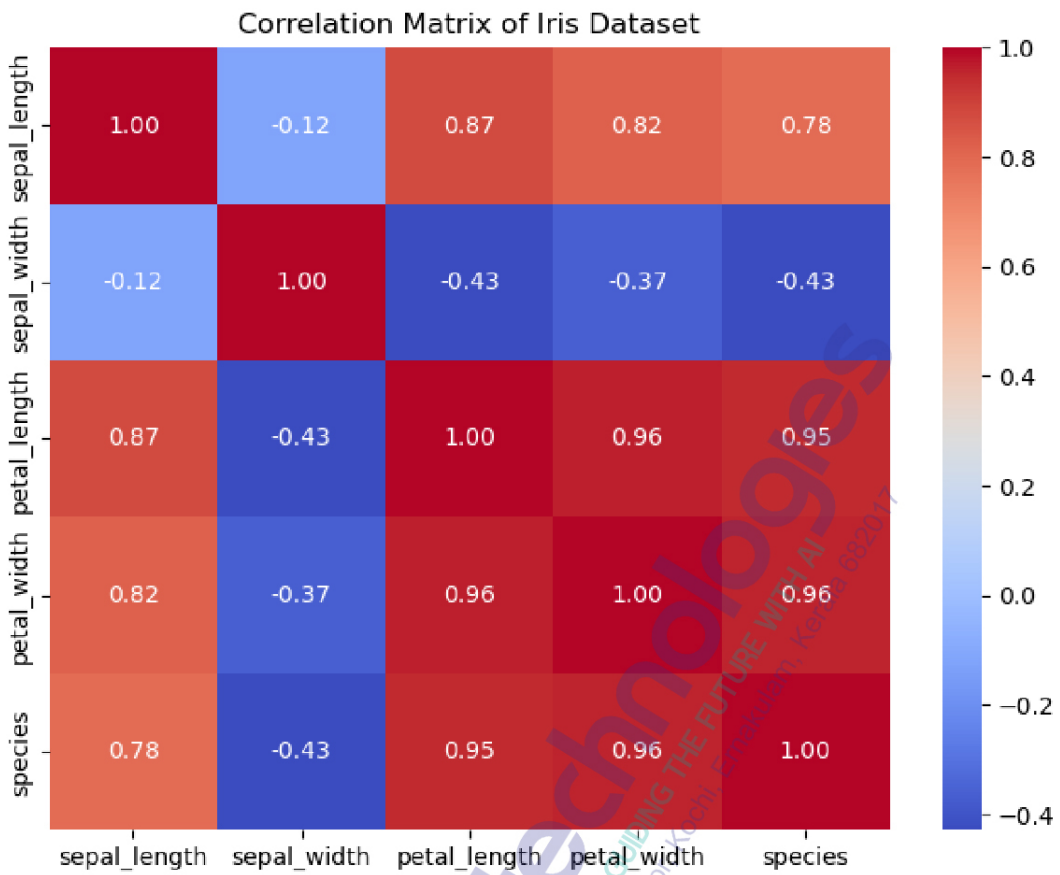
Output:




```
Correlation with species:
sepal_length     0.782561
sepal_width     -0.426658
petal_length     0.949035
petal_width      0.956547
species          1.000000

Name: species, dtype: float64
```

Interpretation: Petal length and petal width shows a correlation of 96% that means multicollinearity is existing, so ignore any of the two. Comparing the

variables with the output variables Petal length/Petal width and Sepal length is having good correlation with the output species that means these two variables dominates the prediction of the output. And these two variables can treated as features.

Extracting dependent (targets) and independent variables (features) with indexing

In machine learning, it is important to distinguish the matrix of features (independent variables) and dependent variables from dataset. In our dataset, there are three independent variables that are **Country, Age**, and **Salary**, and one is a dependent variable which is **Purchased**.

We can select using loc, iloc functions, selected columns

Method 1: Using iloc method

```
import pandas as pd
df=pd.read_csv(r"E:\DataScience\Datasets\ml\Data_2.csv")
# Full dataset
print(df)
print("Independent variables")
x=df.iloc[: , :-1].values
print(x)
print("Dependent variable")
y= df.iloc[:,3].values
print(y)
```

Method 2: Using loc method

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```
import pandas as pd
df=pd.read_csv(r" E:\DataScience\Datasets\ml\Data_2.csv")
# Full dataset
print(df)
print("Independent variables")
x=df.loc[: , ['Country','Age','Salary']].values
print(x)
print("Dependent variable")
y= df.loc[:,'Purchased'].values
print(y)
```

Method 3: Direct DataFrame filtering based on column names

```
import pandas as pd
df=pd.read_csv(r" E:\DataScience\Datasets\ml\Data_2.csv")
# Full dataset
print(df)
print("Independent variables")
x=df[['Country','Age','Salary']].values
print(x)
print("Dependent variable")
y= df['Purchased'].values
print(y)
```

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Output:

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40.0	NaN	Yes
5	France	35.0	58000.0	Yes
6	Spain	NaN	52000.0	No
7	France	48.0	79000.0	Yes
8	Germany	50.0	83000.0	No
9	France	37.0	67000.0	Yes

Independent variables

```
[['France' 44.0 72000.0]
 ['Spain' 27.0 48000.0]
 ['Germany' 30.0 54000.0]
 ['Spain' 38.0 61000.0]
 ['Germany' 40.0 nan]
 ['France' 35.0 58000.0]
```

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```
 ['Spain' nan 52000.0]
 ['France' 48.0 79000.0]
 ['Germany' 50.0 83000.0]
 ['France' 37.0 67000.0]]
Dependent variable
['No' 'Yes' 'No' 'No' 'Yes' 'Yes' 'No' 'Yes' 'No' 'Yes']
```

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